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Advanced Computer Architecture

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Problem

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
I1: LD F6 32+ R2	1	2	7	8
I2: ADDD F2 F6 F4	2	9	11	12
I3: MULTD F0 F12 F2	3	13	43	44
I4: SUBD F12 F2 F6	4	13	15	45
I5: ADDD F0 F12 F2	45	46	48	49

A. List all the possible conflicts in the code.

B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

C. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

ANSWERS